SFEM 2026

# AI-Driven Climate-Resilient Adaptation Model of Korean Forests (CRAFT)

## Grid-scale Integration of Growth, Carbon, Habitat, and Disaster

Heedo Choi[1*] · Jee Soo Park[1] · Chul-Hee Lim[1,2,3†]

[1] Department of Climate Technology Convergence, Kookmin University
[2] Department of Forestry, Environment and Systems, Kookmin University
[3] Department of Forest Resources, Kookmin University

# Contents

# 01 Background and objective

What is happening to Korean forests, and why the three belong together

Each of the three functions is already well modelled in Korea. What is missing is a single frame in which they constrain one another.

## In this section

### The gap

Carbon, habitat and disaster are estimated by separate models built at different scales, so the effect of one on another cannot be traced.

### The requirement

A shared spatial unit and a shared calendar, fine enough to resolve a stand and wide enough to cover the country.

### The claim

One grid carries the three functions, and one annual loop moves them forward together rather than side by side.

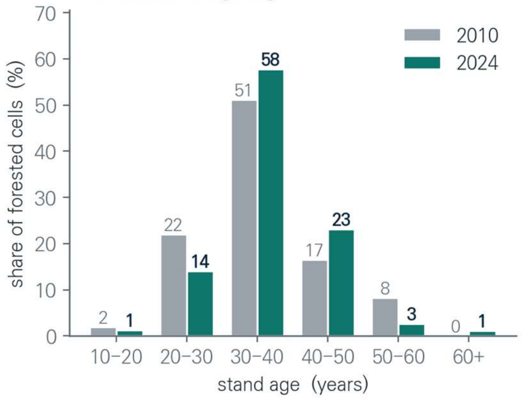
Where we are

covered on slides 4 and 5

# Korean forests are changing

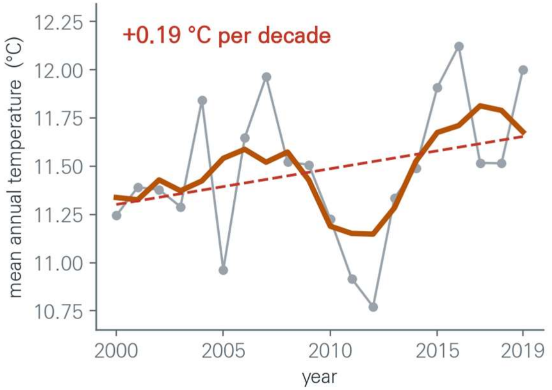
Almost the whole of the Korean forest was replanted in the decades after the Korean War, and it is now passing through the ages at which a stand accumulates carbon most rapidly.
Over the same period the climate has warmed and the largest fire seasons in the record have occurred.

### The forest is ageing



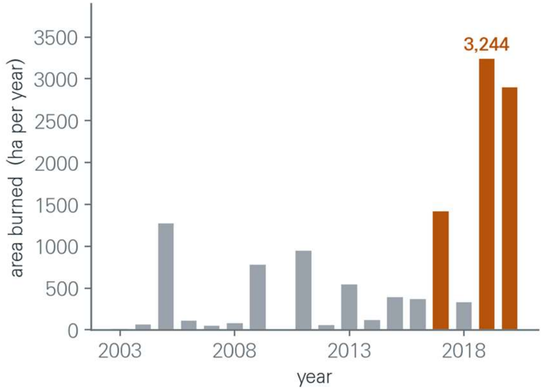

Almost every stand was planted after the 1950s, and the age class that holds the most carbon is now the one that is filling up.

### The climate is warming




Averaged over every forested cell in the country. The thick line is a five year mean, the dashed line the linear trend.

### Disturbance is taking more back




Two of the three worst fire years on record fall at the end of the series.
A single season can remove carbon that took decades to build.

## Forest age

In 2010 half of the forested cells carried stands of thirty to forty years.
By 2024 the next class had almost doubled.

## Climate

Temperature over forested land rose by about 0.19 degrees each decade, and growth, mortality and fire weather respond to it.

## Disturbance

A single season can remove carbon that took decades to accumulate, as 2019 and 2020 showed.

# Three functions, three models

Korean forests are expected to store carbon, provide habitat for major species, and withstand wildfires and landslides. Each function has well-established models, but they are developed independently on different spatial units and reference years.

| Carbon | Habitat | Disaster |
|---|---|---|
| national inventory scale | species distribution models | event-based risk maps |
| *$tC\ ha^{-1}$* | *suitability 0–1* | *probability* |

*different grids · different years · different species assumptions*

**This study estimates all three on one 1 km grid, then lets them interact year by year.**

**Key point** The objective is not a better single model, but a common frame in which the three can interact and be evaluated together.

# 02 Framework and data

Four layers, one national grid, calibrated on the inventory

Everything downstream rests on one decision: fix the spatial unit first, calibrate once, and then leave the parameters alone.

## In this section

### Four layers

A one-time calibration, three trained modules, an annual simulation loop, and a spatial correction network above them.

### One grid

Cells of one kilometre covering the forested area of the Republic of Korea, reconstructed at four observed timepoints.

### Five sources

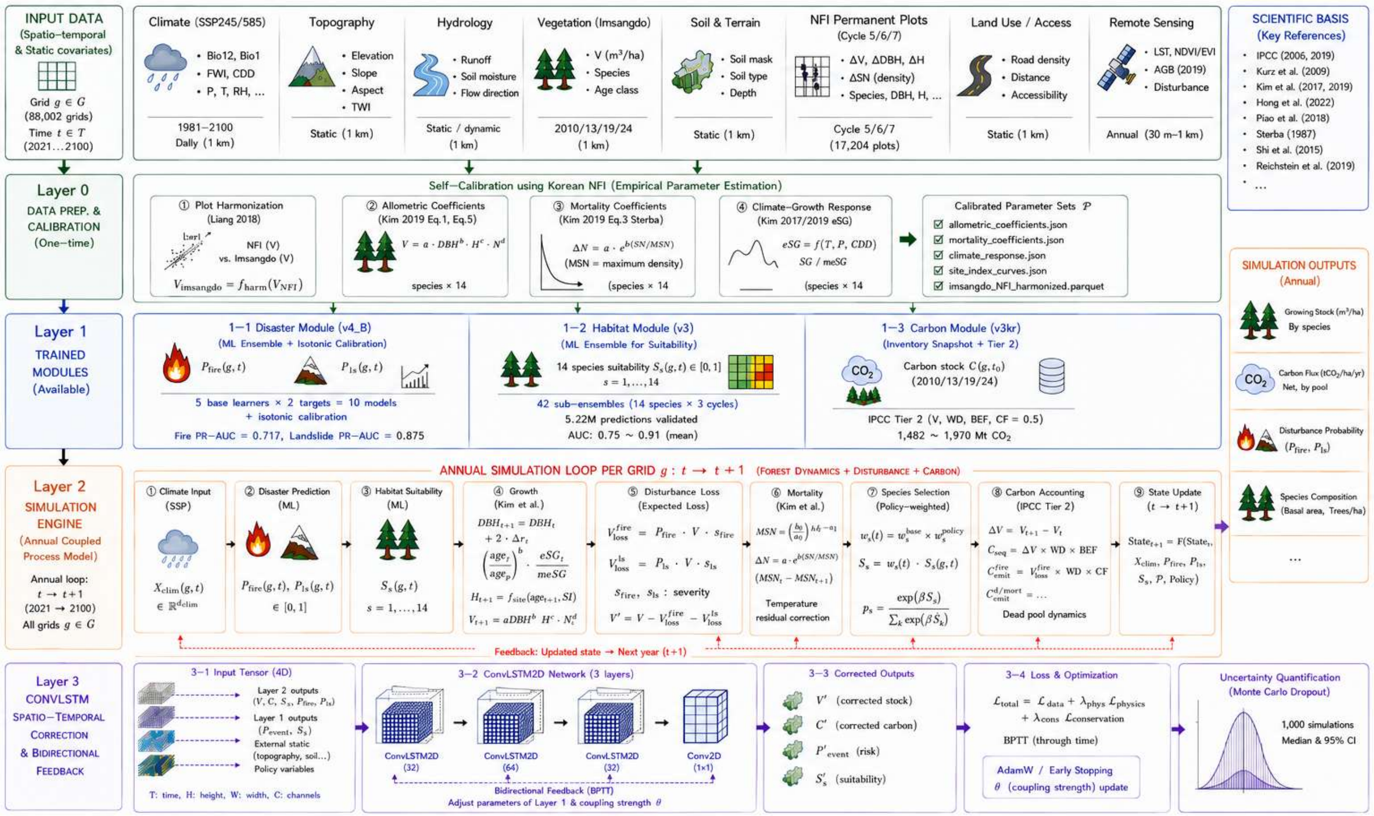
Inventory plots, the forest-type map, certified carbon coefficients, organ-level allometry, and daily climate records.

Where we are

covered on slides 7 to 10

# CRAFT
# (AI-Driven Climate-Resilient Adaptation Model of Korean Forests)

The whole framework on one page. Input data feed a one-time calibration, which supplies parameters to three trained modules; those modules are then coupled in an annual simulation loop and finally corrected by a spatiotemporal network.




## Input data

Three inventory cycles, four map editions, certified coefficients and climate records.

## Layer 0 · Calibration

Four parameter sets estimated once on the national inventory.

## Layer 1 · Modules

Carbon by a process-based chain; habitat and disaster by ensembles.

## Layer 2 · Annual loop

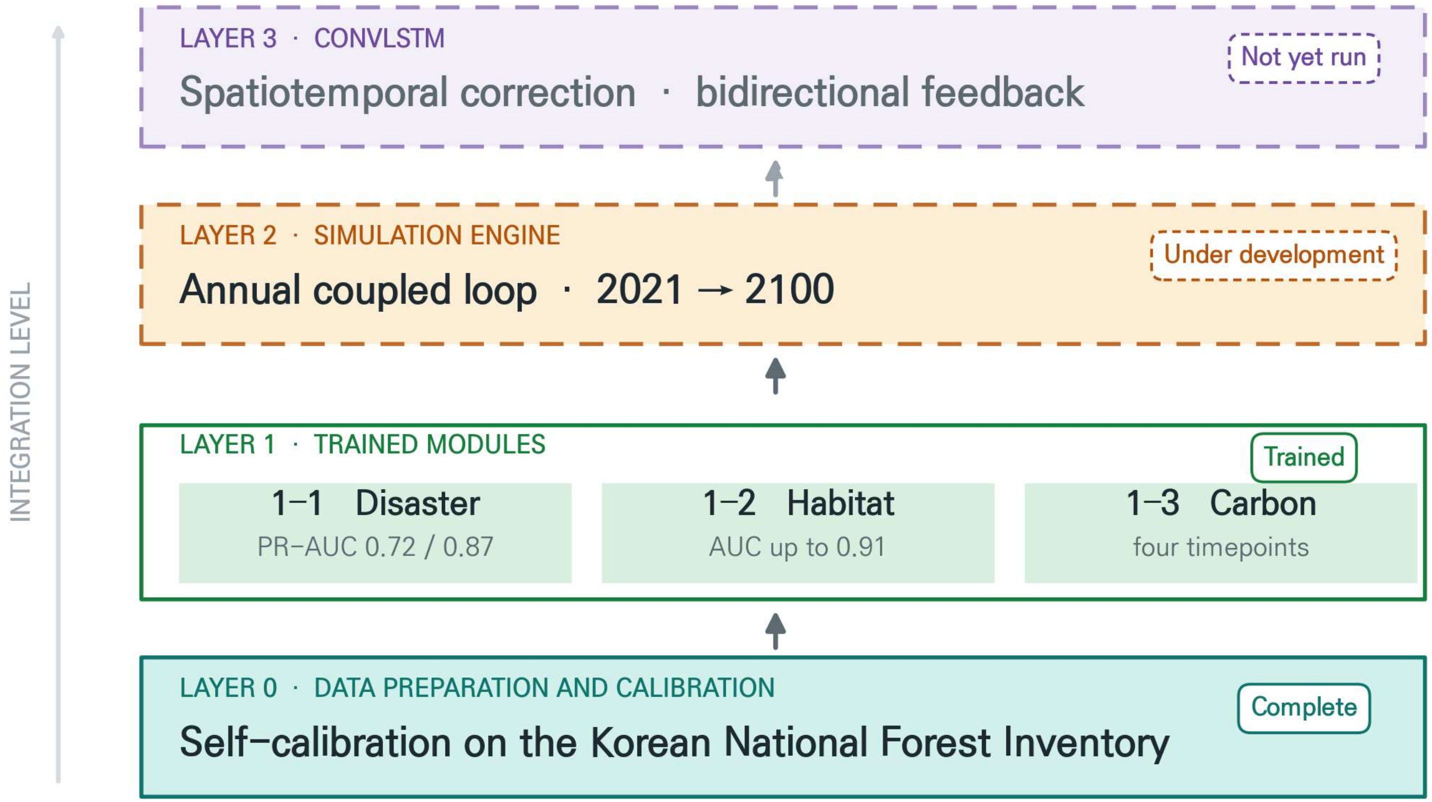
Coupled and stepped forward one year at a time, now run through to 2100.

## Layer 3 · Correction

A recurrent network learning space and time together. Designed, not yet run.

# CRAFT framework

Layer 0 calibrates parameters on the national inventory. Layer 1 holds the three trained modules. Layer 2 couples them into an annual loop, validated on the historical period and now run through to 2100. Layer 3 is designed but not yet run.




### Calibration

The quantities linking the map to the inventory are estimated once, so a later change in carbon is a change in the forest.

### Module fitting

Each module is fitted and validated against its own target before any coupling is attempted.

### Coupling

Only Layer 2 allows the modules to affect one another, and only through the variables the engine updates each year.

### Implementation status

The first three layers are complete and validated. The spatial correction above them is designed but not yet fitted.

# Study area and grid



The analysis covers the entire forested area of the Republic of Korea. The country is divided into one–kilometre grid cells, and each cell is reconstructed at four timepoints for which a national forest–type map exists.

| 1 km × 1 km | 4 timepoints | 14 major species |
| --- | --- | --- |
| about 88,000 forested cells | 2010 · 2013 · 2019 · 2024 | kept as a mixture per cell |

CONVENTIONAL

SPECIES COMPOSITION REFLECTED


One dominant species per cell

mixed stands are lost

Every cell keeps its species mix

fraction–weighted estimation

Threshold 5% · remainder by broad class

**Cells kept** a cell enters the analysis when elevation is present and forest cover is above zero. No–data and non–forest cells are removed before any training.

## Why the mixture is kept

More than half of Korean grid cells hold several species in meaningful proportions. Reducing a cell to whichever species dominates discards that structure, and with it the differences in wood density and expansion that decide how much carbon it holds.

Species below five percent are folded into a broad conifer or broadleaf class, so no forested area is dropped.

# Data integration

Five independent sources are brought onto the common 1 km grid. Three of them are national standards, which keeps the carbon results comparable with national inventory reporting.

INPUT DATA

**National Forest Inventory**
5th · 6th · 7th cycles · 44,928 plots

**National forest–type map**
2010 · 2013 · 2019 · 2024 · 14 species

**Allometric equations**
NIFoS · 22 species × 4 organs

**Certified carbon factors**
NIR–GIR · density, BEF, root–to–shoot

**Climate · terrain · records**
daily climate, DEM, disaster history

**1 km × 1 km**
national grid
☐ 102,800 cells

PREDICTION MODULES

**Carbon**
stand volume → carbon stock

**Habitat**
suitability for 14 species

**Disaster**
wildfire · landslide

**Why the source list matters**
Three of the five are national standards, so the carbon result can be placed beside national reporting rather than read as a number internal to this model.

**National Forest Inventory**
44,928 plots from cycles five, six and seven, with tree–level remeasurement.

**Forest–type map**
Four editions, 2010 to 2024, read as species fractions per cell.

**Certified coefficients**
Wood density, biomass expansion and root ratios from national reporting.

**Allometric equations**
Organ–level equations for 22 species, national research institute.

**Climate and records**
Daily climate, fire weather indices, and recorded fire and landslide history.

# 03 Layer 1 · Three trained modules

Disaster, habitat and carbon with their validation

Each module is trained separately and then frozen, so a later change in carbon is a change in the forest and not in the fit.

## In this section

### Machine learning modules

Habitat and disaster follow one ensemble design: screened predictors, five learners, spatial blocks reserved for validation, and probabilities calibrated afterwards.

### Process based module

Carbon is derived step by step from stand age to volume to carbon, with published equation forms and nationally certified conversion factors.

### Calibration and validation

Comparison with published national statistics revealed a twofold overstatement in the earlier chain. Closing it is the largest single result in this section.

Where we are

covered on slides 12 to 21

# Layer 0 · Calibration

Before any module is trained, the quantities that connect the national forest–type map to the inventory are estimated once and then frozen. Everything above this layer reads the same table, so a later difference is a difference in the forest and not in the equation.

①
**Plot harmonisation**
inventory volume aligned with the official figure

②
**Height conversion**
map height converted to dominant and mean height

③
**Density lookup**
density read from inventory medians on map classes

④
**Growth and volume**
diameter and volume coefficients refitted

⑤
**Scale transfer**
one factor per species, plot support to grid

Every parameter set is written once, on the same basis it will later be read on, and is then reused by every layer above.

## What is fitted here

Height conversion ratios, the density lookup, the diameter and volume coefficients, and the plot–to–grid scale factor.

## The rule this imposes

A coefficient has to be read on the same basis it was fitted on. Section 03 shows what happens when that rule is broken.

# Habitat and disaster modules

Both machine learning modules follow the ensemble design of biomod2 (Thuiller et al. 2009), reimplemented in Python so that the same procedure applies to species occurrence and to hazard occurrence.

### 1 Variable screening

Spearman |r| 〉0.7

VIF 〉10

30 → 18 habitat
83 → 60 wildfire
65 → 47 landslide

### 2 Base learners

- GLM
- Random forest
- Gradient boosting
- XGBoost
- MaxEnt · LightGBM

### 3 Spatial block CV



KMeans 10 blocks

4 folds × 3 runs

**60 sub-models per target**

### 4 Ensemble

weight $\propto$ ( PR−AUC − base )$^2$

quality filter

TSS 〉0.5
AUC 〉0.7

### Background sampling

Habitat target−group absence from inventory plots

Disaster 1 : 1 sampling from forested cells only

### Calibration (disaster module)

Isotonic regression on out−of−fold predictions,

then Youden J threshold.

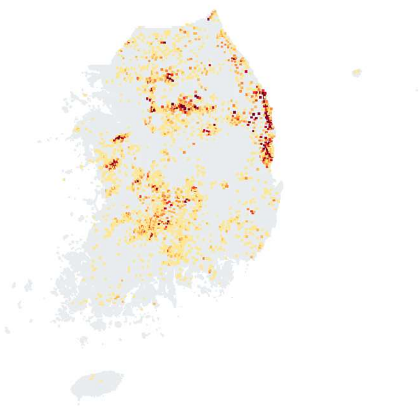
| | | | |
|---|---|---|---|
| **83 → 60** wildfire predictors | **65 → 47** landslide predictors | **30 → 18** habitat predictors | **12** spatial splits per target |

# Module 1–1 · Disaster

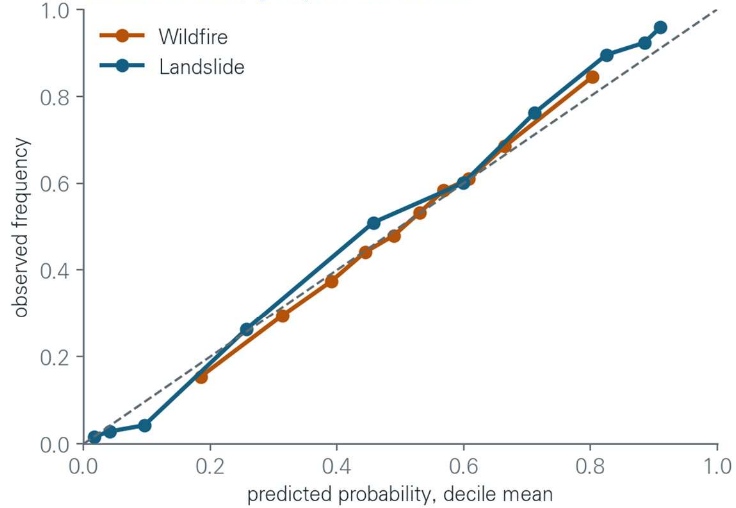
The wildfire register used here covers 2000 to 2019 and the landslide register 2010 to 2019. Within those windows the module estimates the annual probability that a cell burns or slides, from climate, terrain, forest condition and human access. The two hazards are fitted separately because their drivers differ.

Wildfire
10,819 cell years used, 2000 to 2019

Landslide
2,111 cell years used, 2010 to 2019



Prediction against outcome
validation cells, grouped into deciles

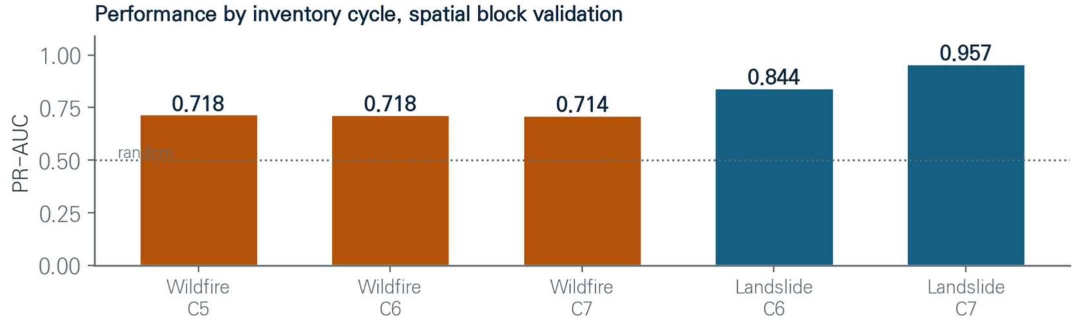


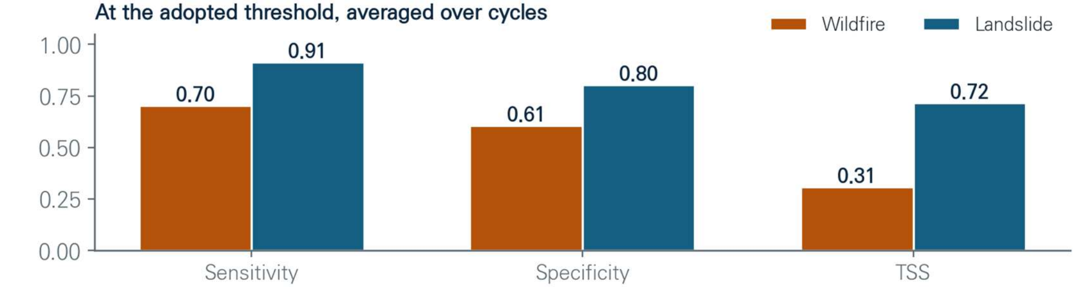

| | | | |
|---|---|---|---|
| **10,819** wildfire cell years | **2,111** landslide cell years | **60 and 47** predictors retained | **5** learners per hazard |

**Key point** Every recorded event is matched with a forested cell that did not burn in the same year, so the two classes differ only in outcome.

# Module 1–1 · Disaster performance

Performance is reported on spatial blocks that were set aside during fitting, and separately for each inventory cycle so that a good score cannot arise from one favourable period.
Wildfire is stable across all three cycles, and landslide improves as the register becomes more complete.




### Reading the lower panel

Most events are captured, and many cells that do not burn are also flagged.

## Probability calibration

A model can order cells correctly and still report the wrong level. Isotonic regression on predictions made outside the fitting fold reduces the calibration error from about 0.13 to 0.02, and leaves the ordering unchanged.

## Ensemble weighting

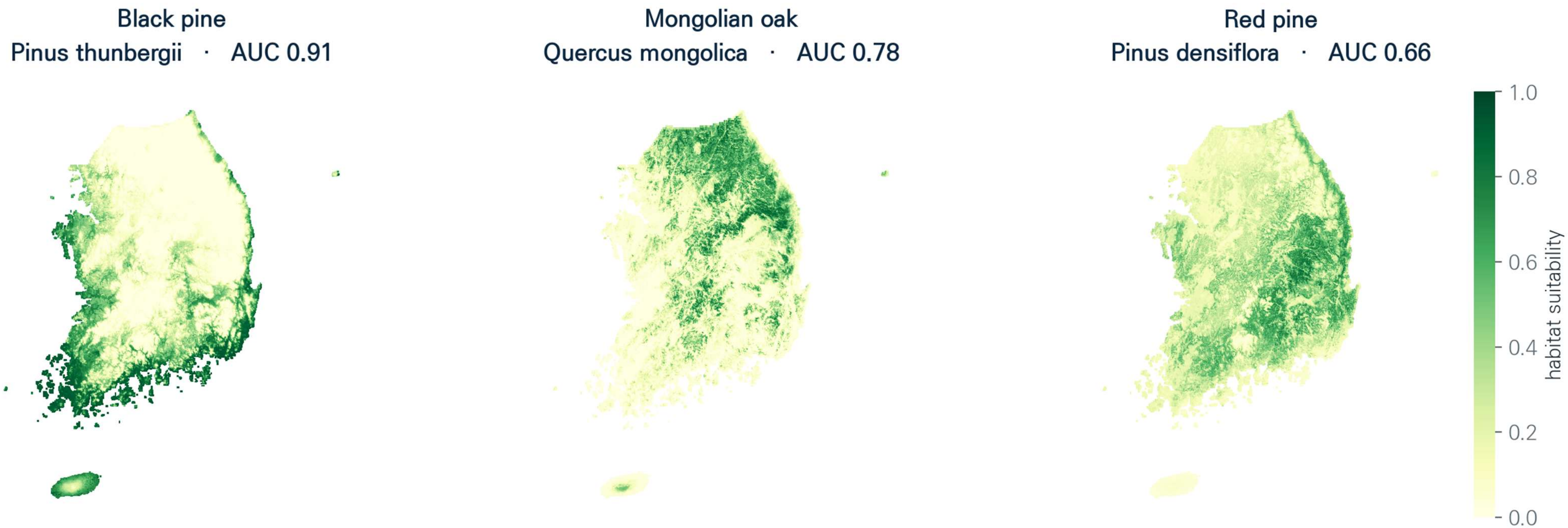
Five learners are fitted for each hazard and combined in proportion to squared skill. Choosing a single best learner on validation data would amount to a second fitting step.

## What the score measures

The score describes how well cells are ordered within a year. How often a fire occurs is a separate quantity, and it is set in Layer 2.

# Module 1–2 · Habitat

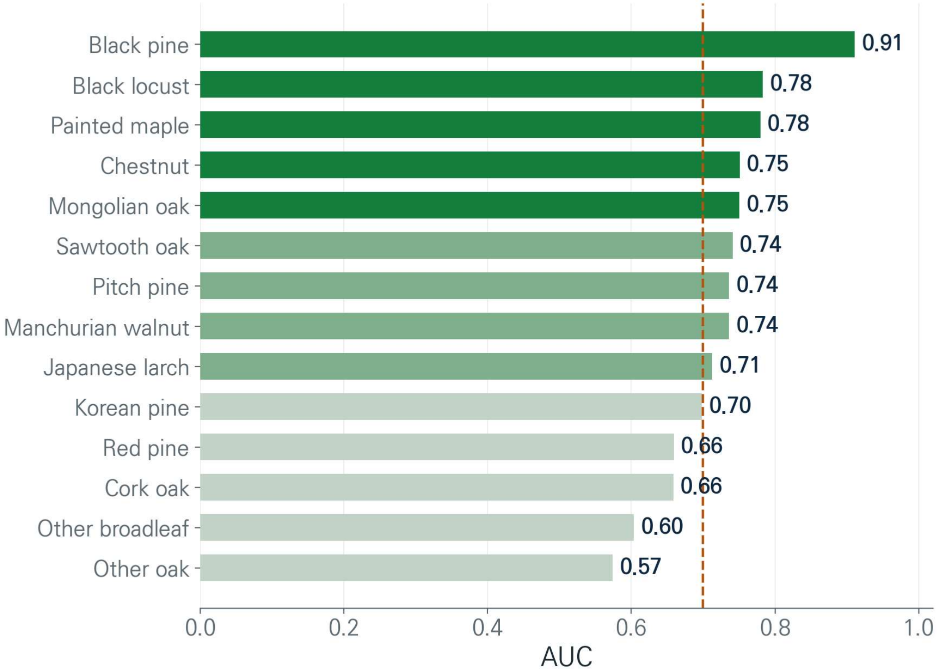
For each of fourteen tree species the module estimates how suitable every cell is, from climate, terrain and soil, with each inventory cycle trained on its own climate window.
How well a species can be predicted turns out to depend on how sharply its range is bounded, not on how common it is.




| | | | |
|---|---|---|---|
| **0.91** best species, black pine | **9 of 14** above AUC 0.70 | **2,515** sub-models trained | **± 5 years** climate window per plot |

# Module 1–2 · Habitat performance

Each species is fitted on its own and enters the ensemble only when it satisfies two criteria at the same time.
Nine of the fourteen species satisfy both.
The remaining five are retained in the composition and treated as background rather than removed.




## Inclusion criteria

A sub model is admitted when the true skill statistic exceeds 0.5 and the area under the curve exceeds 0.7. About eighteen percent of the habitat sub models satisfy both, and the rest are discarded rather than down weighted.

## Absence data

Absence is taken from inventory plots that were surveyed and did not record the species, following the target group approach of Phillips et al. 2009, so the contrast is between two measured states.

**Key point** Suitability is a relative weight for one species and not the share of the cell it occupies, so the fourteen values do not sum to one.

# Module 1–3 · Carbon

The carbon module is a process chain rather than a fitted predictor. Stand age is read from the forest type map, and each step takes the output of the previous one until standing volume is converted to carbon by nationally certified factors. The equation forms are published; only their coefficients are re-estimated on Korean inventory plots.

| A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|
| Stand age | Site index | Stand density | Mean height | Mean diameter | Stand volume | Carbon |
| $t$ | $SI = h_o / f(t)$ | $N$ | $h_m = r_{sp} H_{map}$ | $D = a\,e^{b/t} SI^{c} N^{d}$ | $V = \lambda_{sp} a D^{b} h_m^{c} N^{d}$ | $C = V\rho\,BEF(1+R)CF$ |
| forest type map | Chapman and Richards | inventory medians | fitted ratio | Piao et al. 2018 | Kim et al. 2022, Eq. 1 | Kim et al. 2022, Eq. 5 |

$h_o$ dominant height · $H_{map}$ height from the forest type map · $\rho$ wood density · $BEF$ biomass expansion factor · $R$ root to shoot ratio · $CF$ carbon fraction

Every step is evaluated for each species present in a cell and combined by area fraction. $\lambda_{sp}$ carries the volume equation from plot support to grid support.

### Published forms

The diameter and volume equations follow established Korean growth and yield work.

### Certified conversion

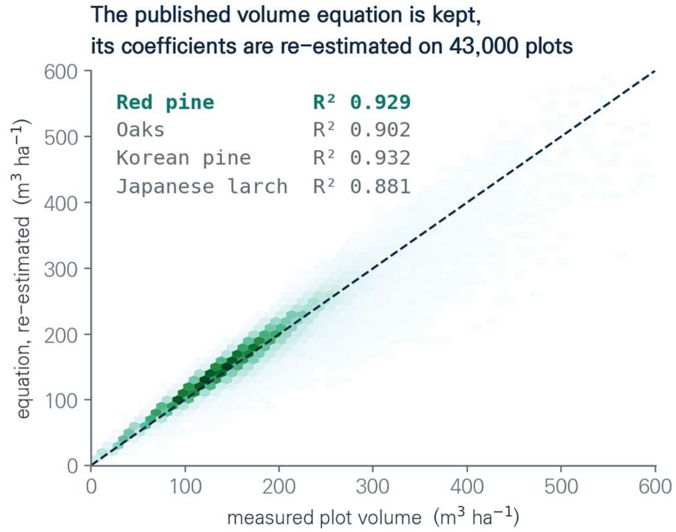
Wood density, expansion and root ratios come from national reporting.

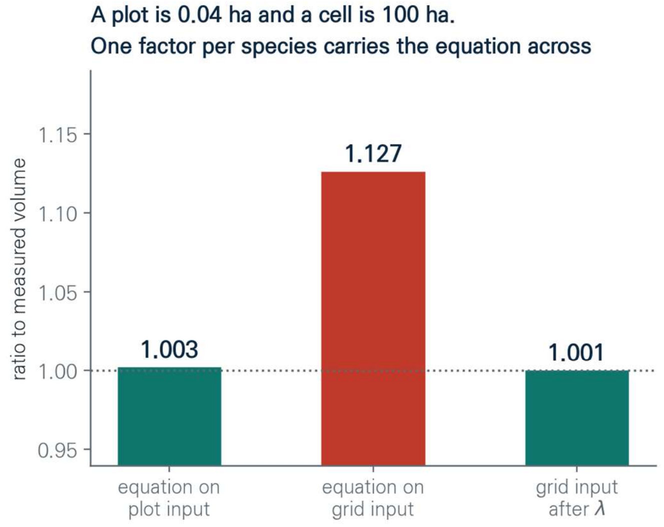
### Species mixture

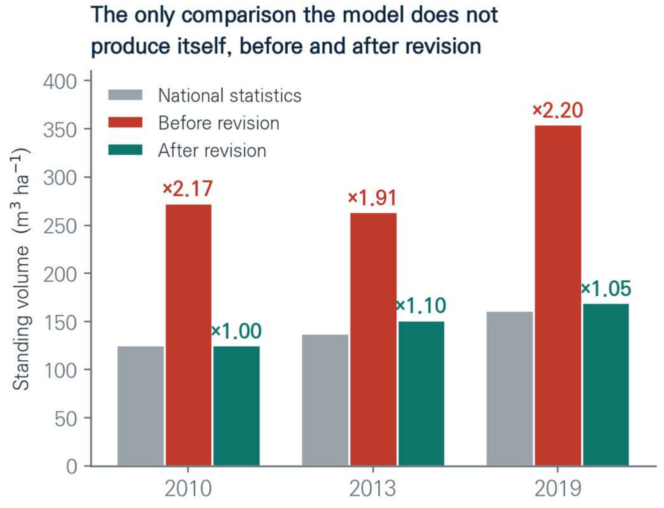
Each step is evaluated for every species in a cell and combined by area fraction.

# Module 1–3 · Carbon calibration

Coefficients are estimated on 43,000 inventory plots of four hundred square metres, and then applied to grid cells of one hundred hectares. The average survives that change of scale but the spread does not, so one factor per species carries the equation across before anything is compared.




### Coefficient estimation

The volume equation is re–estimated on inventory plots, reaching $R^2$ of 0.88 to 0.93.

### Scale correction

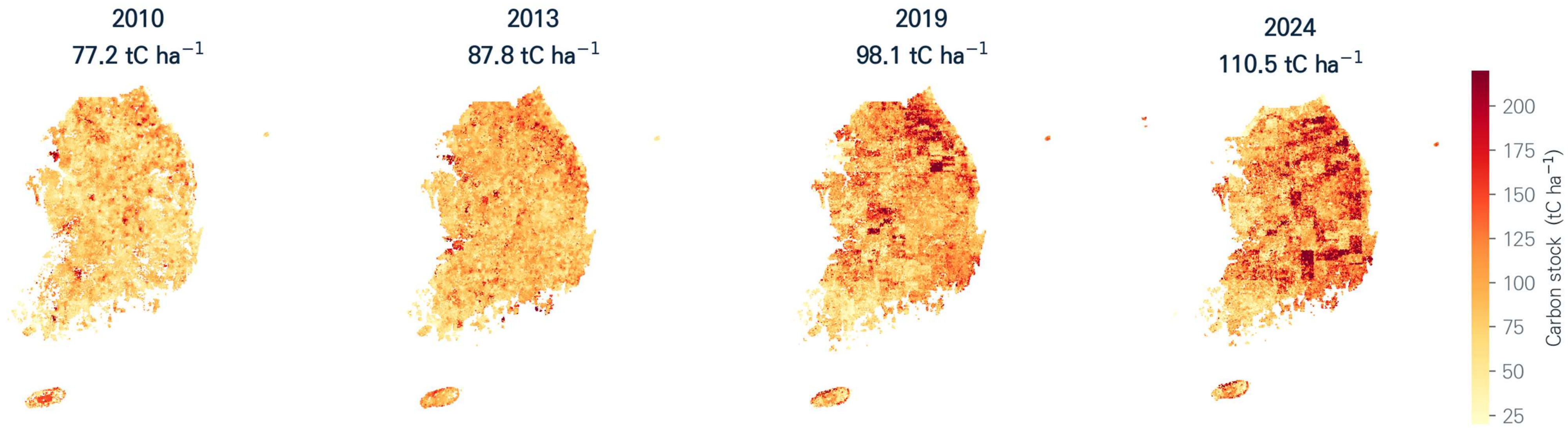
The equation is exact on plot input and reads 12.7 percent high on grid input. One factor per species removes that gap.

### External comparison

Against published national statistics the chain now stands between 1.00 and 1.10, where it previously stood at 2.2.

# Module 1–3 · Carbon results

Grid mean carbon rises from 77 to 110 tonnes per hectare between 2010 and 2024. Every state variable behind that number moves in the direction a maturing and self thinning stand would move, and the reconstructed volume stands within ten percent of published national statistics at every date that can be compared.




2010 uses the first forest-type map edition, whose coverage is smaller (63,854 cells)

| | | | | | |
|---|---|---|---|---|---|
| 35.5 → 36.6 age, yr | 1,049 → 1,112 density, $ha^{-1}$ | 11.1 → 12.3 height, m | 15.3 → 16.8 diameter, cm | 126 → 191 volume, $m^3\ ha^{-1}$ | 77 → 110 carbon, $tC\ ha^{-1}$ |

**Key point** The increase is national forest maturation resolved cell by cell.

# Three modules on one grid

Because the three modules share a spatial unit and a calendar, every grid cell simultaneously carries a carbon stock, a suitability value for each of the fourteen species, and an annual probability for each hazard.

## $R^2$ 0.88–0.93

### Carbon

stand volume reproduction

A process chain from age to carbon using certified coefficients, checked against national statistics.

## AUC 0.91

### Habitat

black pine, best of 14 species

Ensemble species distribution models with a climate window matched to each plot. Nine of fourteen exceed 0.70.

## PR–AUC 0.72 / 0.87

### Disaster

wildfire / landslide

Five learners per hazard, calibrated so the output reads as a probability, and scored on spatial blocks set aside during fitting.

**Three outputs for the same cell, in the same year, from the same species composition.**

**Key point** This shared spatial unit is what makes the next step possible: letting the three interact through time.

**And note** Carbon uses a process chain; habitat and disaster use ensembles.
They are combined by geography, not by a single fitted model.

# 04 Layer 2 · Annual dynamics

Making the grid move, and testing that motion against remeasured plots

A snapshot becomes a trajectory only when growth, loss and recovery are resolved in the same year for the same cell.

## In this section

### The loop

Nine operations run for every cell in every year, in a fixed order, and the state they leave is the next year's input.

### The processes

Four fitted curves, three mortality terms, organ-level disturbance transfer, decaying dead pools, and gap-driven establishment.

### The results

A test against remeasured plots, a projection read as both a stock and a flux, and an interval decomposed into its sources.


Where we are

covered on slides 23 to 32

01 Background | 02 Framework | 03 Modules | **04 Dynamics** | 05 Next step

# Layer 2 · Coupling the modules

Coupling the three modules is not a matter of calling them in turn. Each one has a different relationship with the state of the forest, and that relationship is what decides where it belongs in the annual loop.

### Carbon module

**Not re-run at all**

Its output is a set of curves and certified coefficients, not a predictor. The engine evaluates volume and carbon directly at every step of the loop.

no predictor call needed

### Habitat module

**Computed before the loop**

All 18 predictors are climate, terrain and soil. None of them describes the stand, so suitability for a given year can be obtained ahead of the run and read in.

0 of 18 predictors are stand variables

### Disaster module

**Evaluated inside the loop**

12 of the 60 predictors are stand variables that the engine rewrites every year, so a static risk table cannot be built in advance. Risk has to be asked again each year.

12 of 60 predictors are stand variables

**What that buys: forest state and wildfire risk are solved together rather than in sequence**

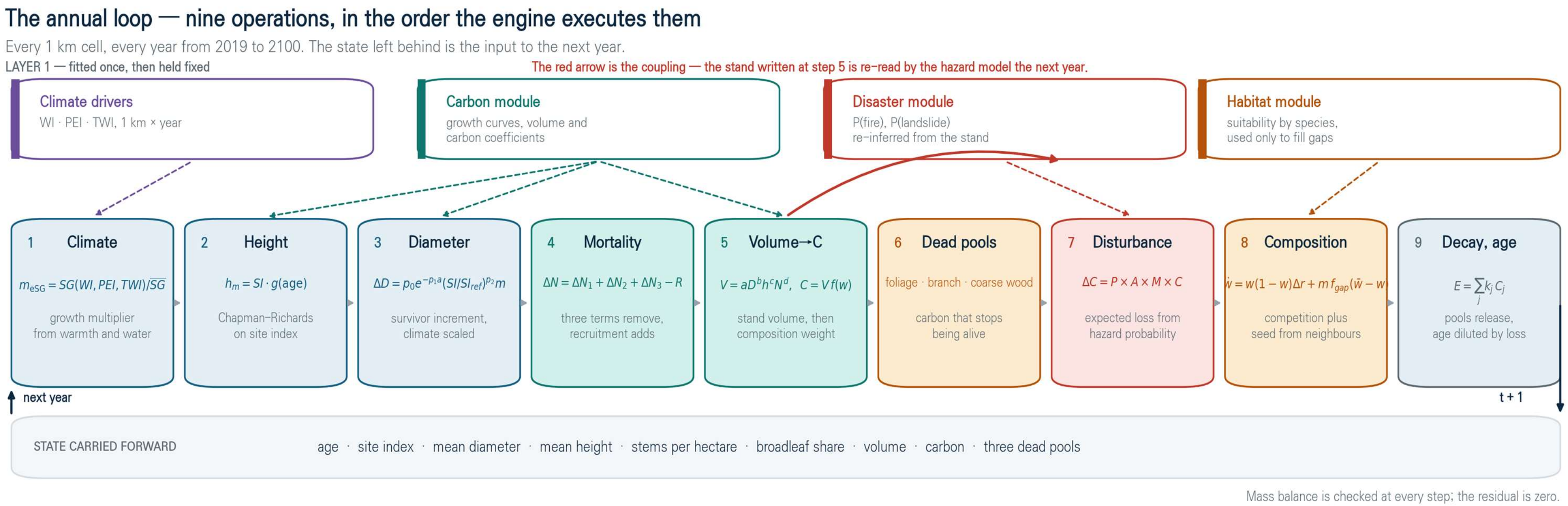
stand matures ▸ fuel load rises ▸ wildfire probability rises ▸ expected loss rises ▸ standing volume falls ▸ fuel load falls

the loop closes here

**Key point** Only the disaster module has to live inside the loop, and that is exactly where the forest and the hazard stop being independent of one another.

# Layer 2 · The annual loop

Layer 1 produces a snapshot. Layer 2 makes it move. Nine operations run for every grid cell in every year, in the order shown, and the state they leave behind is the input to the following year, so the trajectory is generated rather than interpolated between end points.




### Where climate acts

One multiplier, applied to the diameter increment and nowhere else in the loop.

### Where the modules act

Hazard at step seven, suitability at step eight, and at no other point.

### What closes

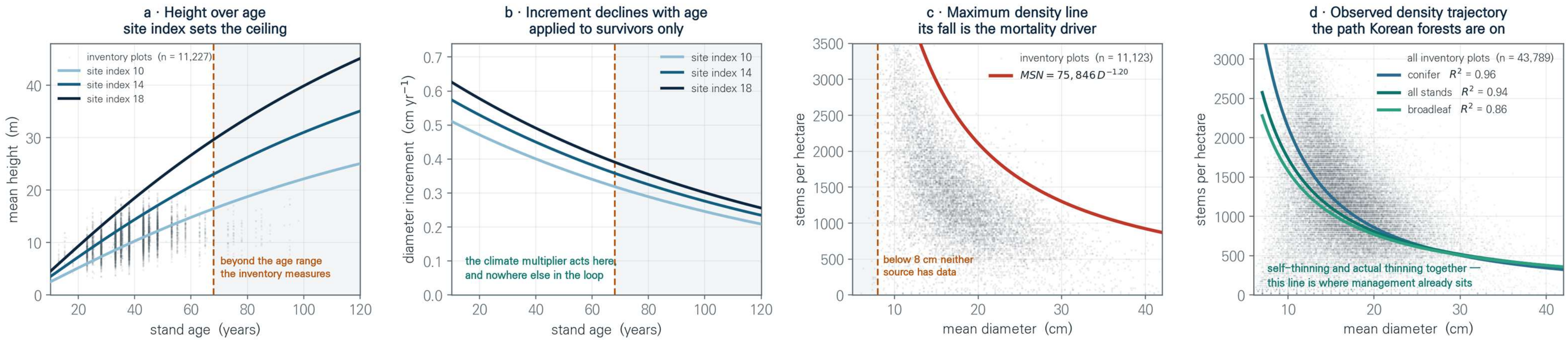
Living and dead carbon are separate accounts whose balance closes to zero.

# Layer 2 · The curves it runs on

Growth and mortality are not prescribed year by year. They fall out of four curves fitted to the national inventory: height over age, the diameter increment, the greatest density a stand of a given size can carry, and the density path Korean forests are actually on.

**The four curves the engine runs on — every year of growth and mortality comes out of these**

Shown for Korean red pine, the most widespread species. Each curve is fitted to the national inventory, and each is drawn against the plots that fitted it.




| | | | |
|---|---|---|---|
| **43,659** inventory plots behind the curves | **$R^2$ 0.94** observed density trajectory | **68 years** the oldest age the inventory reaches | **8 cm** the smallest diameter it records |

**Key point** Because the domain of each curve is recorded, the projection can state where it is reading inside the data and where it is reading beyond it.

# Layer 2 · Mortality

Growth is applied to the trees that survive, so a change in mean diameter is not confused with a change in which trees are present. Mortality is then the sum of three terms, because the published equation on its own responds to how fast density is changing and not to whether it is already too high.

**Mortality is the sum of three terms — and each one acts in a different part of the density range**

The published equation responds to how fast the maximum density line is falling. On its own it cannot return a stand that is already above that line, and it carries no information about how Korean stands are actually managed.

**Density change**

Kim et al. 2017

$$\Delta N_1 = a\, e^{b(SN/MSN - 1)}\, (MSN_t - MSN_{t+1})$$

Trees enlarge, the maximum density line falls, and the drop is the number of stems removed.

$b$ = 1.48 · MSN evaluated at $D \geq 8$ cm

**Level violation**

self-thinning restoration

$$\Delta N_2 = \max(SN - MSN,\ 0) \cdot \lambda_{ST}$$

The first term reads only the change, so a stand already above the line is never brought back.

$\lambda_{ST}$ = 0.35

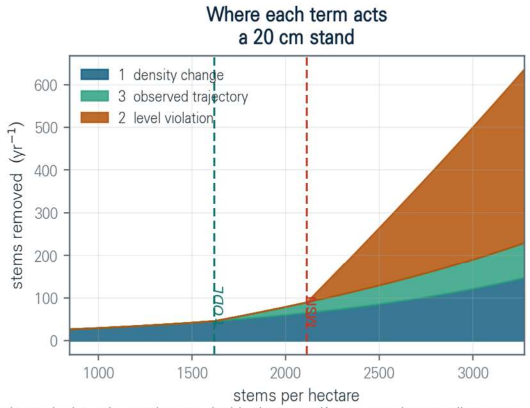
**Observed trajectory**

current management

$$\Delta N_3 = \max(N - \tau\, ODL,\ 0) \cdot \lambda_{ODL}$$

The observed density path already contains Korean thinning, so converging on it states the scenario.

$\tau$ = 2.0 · $\lambda_{ODL}$ = 0.05




The third term is the only one that acts inside the range Korean stands normally occupy.

| | | | |
|---|---|---|---|
| **9,593** remeasured plots behind the calibration | **b = 1.48** density response, red pine | **$R^2$ 0.94** the observed density path | **2 constants** set the two restoration rates |

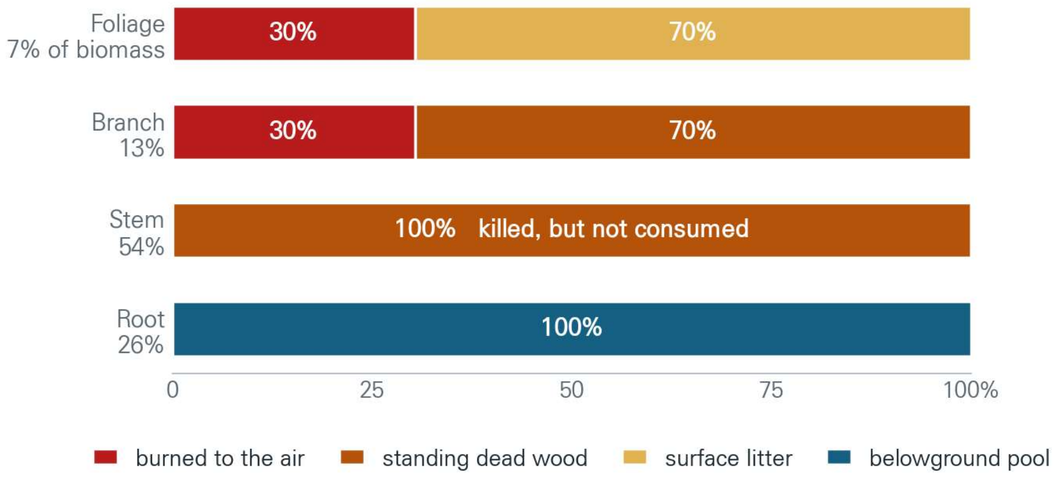
**Key point** The third term is where management enters. The path a stand converges on is the path Korean forests are on, so present thinning practice is carried forward rather than assumed away.

# Layer 2 · Disturbance and dead pools

A fire does not burn a forest to ash, and carbon that stops being alive does not leave in the same year. The loss is resolved organ by organ, each organ is routed to the pool it actually enters, and each pool releases on its own measured decay constant.

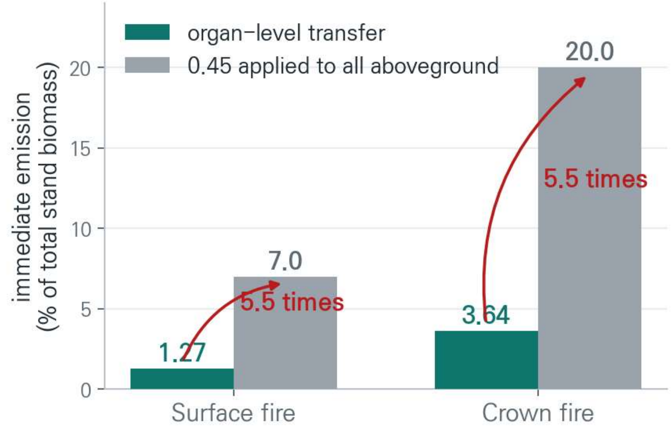
## Where the carbon of a burnt tree goes

Organ−level transfer, applied to the engine carbon so that mass is conserved




## What the shortcut would have cost

Immediate release if the default combustion factor were applied to all biomass

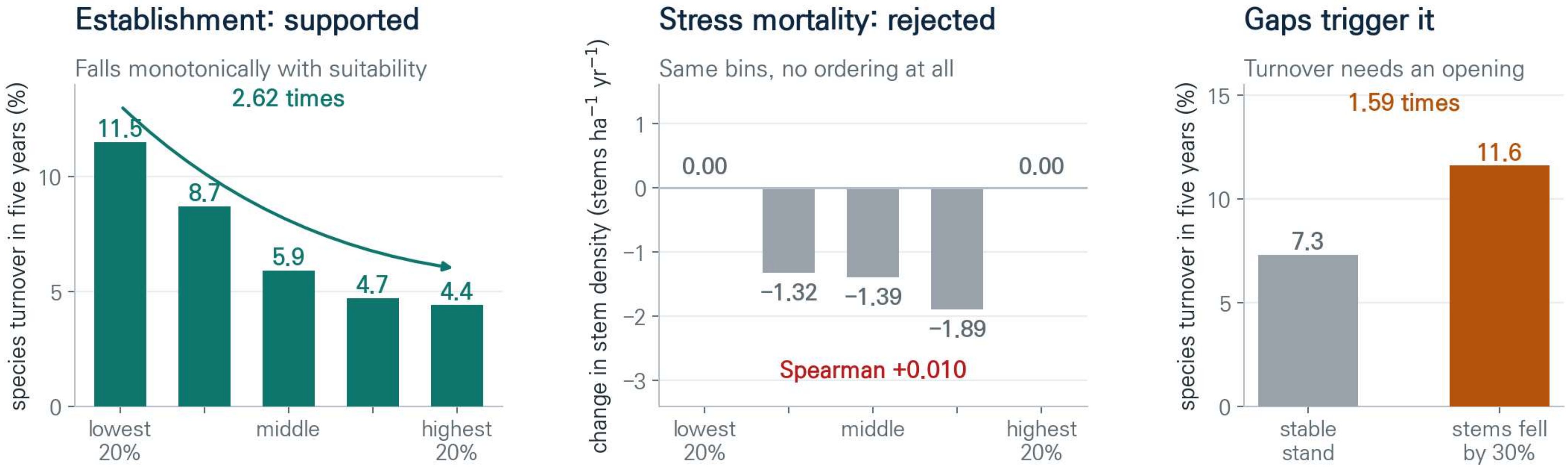



| | | |
|---|---|---|
| **30.4 %** of fine fuel consumed, Hongseong 2023 | **0 %** of living stems burned; they become snags | **5.5 ×** if the default factor were applied to all biomass |

**Key point** In a worked conifer stand, of the eighty−seven tonnes of carbon a fire kills, nine leave that year and seventy−three over the following three decades.

# Layer 2 · Establishment and composition

Trees do not move. Before allowing habitat suitability to act on the forest, we tested on remeasured plots what it can and cannot do. Low suitability does not raise the mortality of standing trees, but a gap opening raises the chance that a different species fills it.



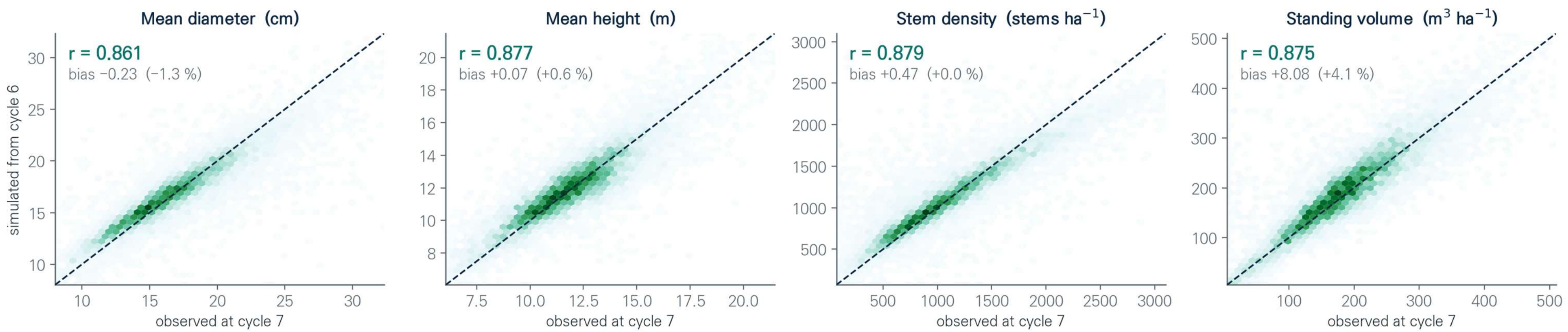

Inventory plots remeasured between the sixth and seventh cycle, n = 7,743. Mature trees survive poor climate for decades, so suitability acts on what fills a gap rather than on what dies. The mortality path was dropped because the data did not support it.

**Key point** Suitability was therefore allowed to decide what fills a gap, and was not allowed to kill a standing tree.

# Layer 2 · Validation

The engine was initialised from the sixth inventory cycle, run forward five years, and compared with the seventh cycle measured at the same plots. This is the only test in the study against data that genuinely moves through time, because the forest–type map is redrawn at each survey.

Run forward five years from inventory cycle 6, compared with the same plots remeasured at cycle 7 · n = 9,593

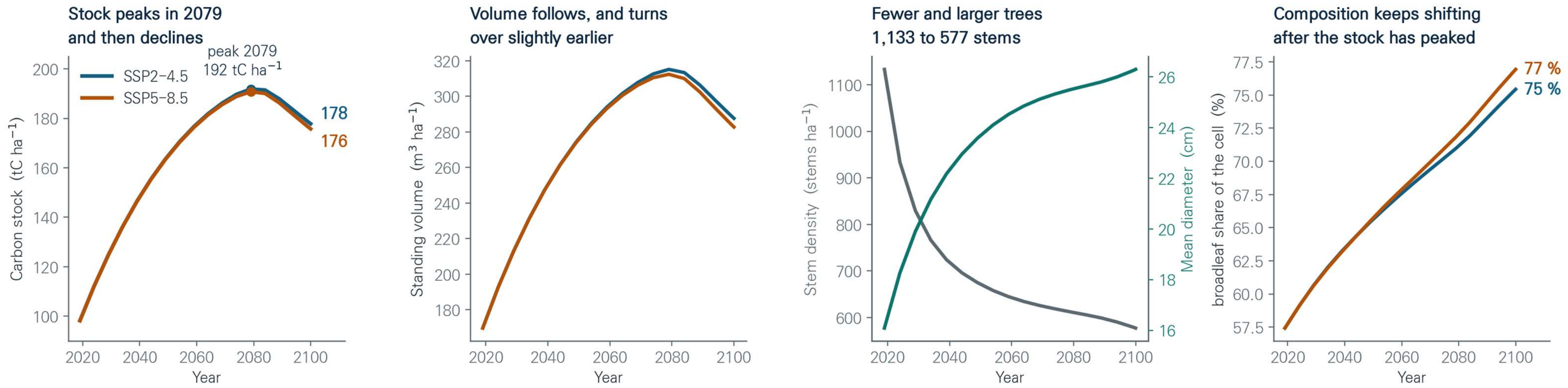



| | | | |
|---|---|---|---|
| **0.86 – 0.88** correlation, four state variables | **9,593** remeasured plots | **+4.1 %** largest bias, standing volume | **0.15** the same engine against the map grid |

**Key point** The engine reproduces remeasured plots at 0.86 to 0.88 and the same engine scores 0.15 against the map grid. The difference is the target, not the model.

# Layer 2 · Projection to 2100

The complete loop was run forward from the 2019 anchor to 2100 under two shared socio–economic pathways, with disaster probability re–inferred from the evolving stand every five years. Carbon stock peaks in 2079 at 192 tonnes per hectare and then declines, while the broadleaf share keeps rising.




| | | | |
|---|---|---|---|
| **98 → 192 → 178** tC ha⁻¹, SSP2–4.5 | **1,133 → 577** stems per hectare | **16.1 → 26.3** mean diameter, cm | **57 → 75 %** broadleaf share |

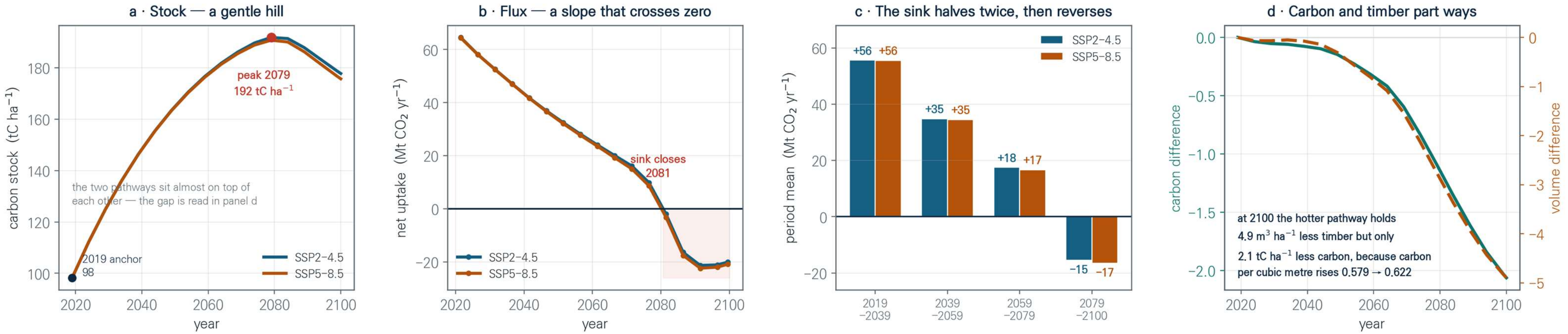
**Key point** Composition keeps changing after the stock has stopped growing, which is the result a stock–only assessment cannot produce.

# Layer 2 · The stock and the flux

A stock and a flux are the same projection read two ways, and they do not say the same thing. The stock is still climbing at mid–century while the annual sink has already halved, and the sink reaches zero around 2080, one year after the stock turns over.

**A stock that is still rising, and a sink that is already closing**

The same projection read two ways. The stock peaks in 2079; the annual sink has already halved by the middle of the century.




**192 tC ha$^{-1}$** peak stock, 2079

**2081** the year net uptake reaches zero

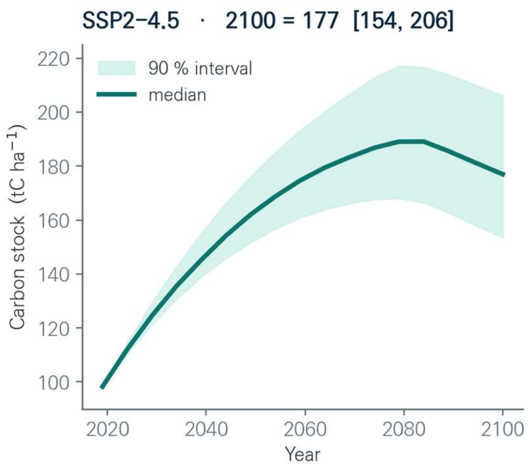
**56 → 18** Mt $CO_2$ yr$^{-1}$, first period to third

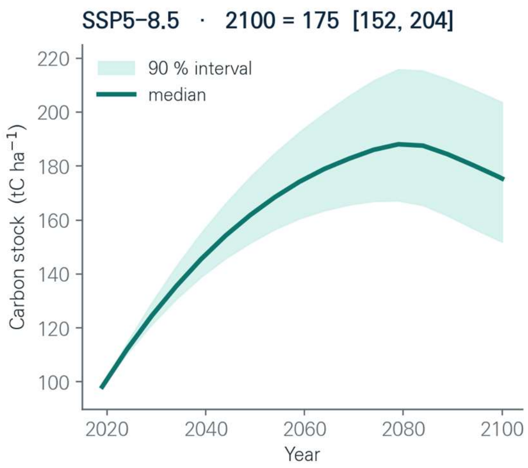
**−15** Mt $CO_2$ yr$^{-1}$ after 2079

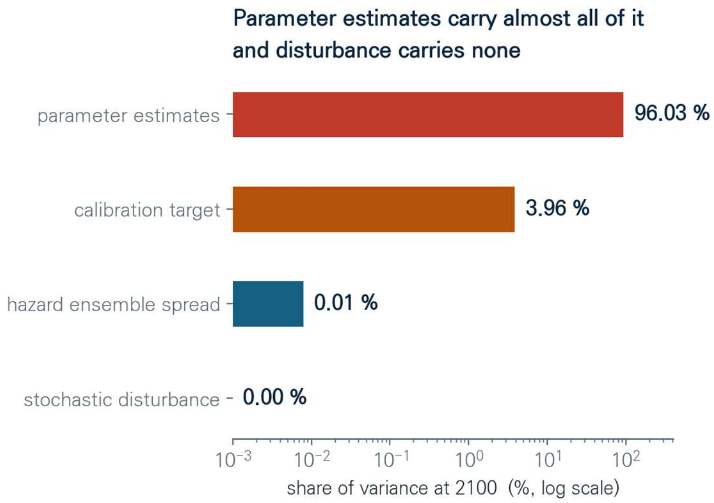
**Key point** A stock target can be met in the same decade the sink that delivers it closes, and the hotter pathway loses more timber than carbon because the forest turns broadleaf.

# Layer 2 · Uncertainty

A projection without an interval is an assertion. Forty realisations were run across four independent sources of uncertainty, and the variance at 2100 was decomposed to find which of them the interval is actually made of.




| | | | |
|---|---|---|---|
| **177 [154, 206]** tC ha$^{-1}$ at 2100, 90 % interval | **96.5 %** from parameter estimates | **4.0 %** from the calibration target | **0.01 %** from hazard and its realisation |

**Key point** Refining the disaster module further would not move the 2100 carbon stock. Narrowing the growth and mortality parameters is the only path that would.

# 05 Next research step

The spatial correction, wood products, and what the framework already gives

The engine is complete, validated and projected. What remains is to let it see across cells, and to let it be harvested.

## In this section

### Spatial correction

A convolutional recurrent network will learn the residual the per–cell engine leaves behind. The design is fixed and the pipeline verified.

### Harvest and wood

A harvest flux and three product pools, which reset stand age and make management a scenario axis in the same units as climate.

### What it already gives

One grid, one calendar and one loop, so carbon, habitat and disturbance constrain each other instead of being reported side by side.

# Layer 3 · Why a spatial correction

Every cell is advanced on its own, so anything that travels across the landscape sits outside the engine by construction. Three processes fall in that category, and each one of them touches a number already shown in this talk.

1. **Stack** — a year becomes one image
2. **Rasterise** — 640 x 612 cells 21 channels
3. **Patch** — spatial blocks held out
4. **ConvLSTM** — encoder and forecaster
5. **Two heads** — occurrence and magnitude
6. **Masked loss** — error terms plus mass balance
7. **Two stages** — fit, then project

**The engine predicts each cell on its own. The network learns what crosses cell boundaries, and corrects the engine rather than replacing it.**

### Input

- 640 x 612 cells at 1 km
- 21 channels over 20 years
- fitted 2000–2019, projected to 2100

### Network

- encoder and forecaster, hidden 32
- two heads, both as residuals
- mass-balance penalty at weight 0.1

### Status

- pipeline verified on a synthetic tile
- spatial blocks and a 2016–2019 hold-out
- not yet trained; no performance claimed

### Wildfire spread

A fire does not stop at a cell boundary, but hazard is estimated for each cell alone.

### Species movement

Broadleaf share rises from 57 to 75 per cent by 2100, but seed has to arrive from somewhere.

### Recovery context

How fast carbon returns after a disturbance depends on the stands around the gap.

# Layer 3 · The model

The correction is a convolutional recurrent network fitted on the CRAFT fields themselves. Each year of the country is one image with many channels, the encoder compresses the observed years into a state, and the forecaster unrolls that state forward and emits a correction field for every year it produces.

### ConvLSTM cell

Every matrix product of a standard LSTM is replaced by a convolution, so the hidden state keeps its spatial layout.

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o)$$

$$H_t = o_t \circ \tanh(C_t)$$

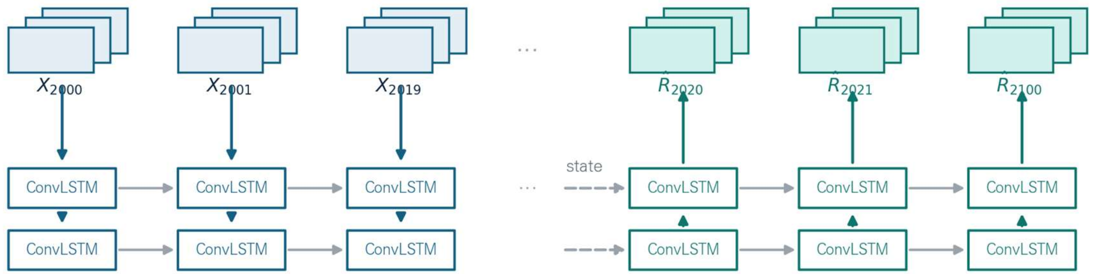
$*$ convolution   $\circ$ elementwise product   $\sigma$ logistic

Shi et al. 2015, Advances in Neural Information Processing Systems

kernel 3 × 3 · hidden channels 32 · two stacked layers · 270,674 parameters

### Encoder and forecaster

The encoder compresses the observed years into a state. The forecaster unrolls that state forward and emits a correction field for every year it produces.




$$\hat{Y}_t = Y_t^{\text{engine}} + \hat{R}_t$$

The network never predicts the state itself. It predicts what the per cell engine could not see, and that residual is added back.

Encoder over 2000 to 2019 · forecaster to 2100 · two heads, occurrence and magnitude · loss adds a carbon mass balance penalty

### What it is given

Twenty years of CRAFT fields on the same 1 km grid, twenty one channels per year.

### What it will correct

The residual between the engine and the observations, added back under a closed carbon balance.

### What follows

Fitting on the real fields, then the same correction applied to both SSP projections.

# Layer 2 · The next axis

The engine has no final harvest, and that is its largest remaining gap. Stand age therefore climbs to 115 years by 2100 and uptake in the 2050s runs at 2.4 times the national projection. Adding a harvest flux requires no new mechanism, only a fourth destination in a transfer matrix that already routes carbon to fire and to landslide.

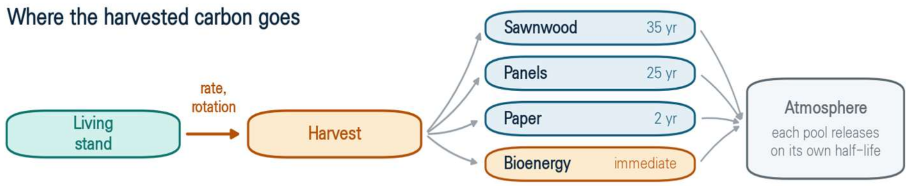
## The next axis — a harvest flux, and what it turns into

Only one new destination is added to a transfer matrix that already routes carbon to fire and to landslide. Every coefficient exists in national statistics.

### The gap it closes

| | |
|---|---|
| 115 yr | mean stand age at 2100, because nothing is ever cut |
| 2.4 × | uptake in the 2050s against the national projection |
| 8.2 % | of remeasured plots already carry a harvest signature |

### Where the harvested carbon goes




half-lives from the 2019 IPCC Refinement · conversion shares from national timber statistics

### What it unlocks

- Harvest intensity and rotation length become scenario axes beside the climate pathway.
- Stand age resets, so the engine can express the management cycle it currently cannot.
- A substitution account becomes possible, which is what a management decision actually turns on.

HOW LAYER 2 GROWS FROM HERE

- **Stand dynamics** ✓ — fitted and validated
- **Disturbance and dead pools** ✓ — organ-level transfer
- **Establishment and composition** ✓ — competition with seed supply
- **Uncertainty** ✓ — forty realisations, four axes
- **Harvest and wood products** → — the axis described above
- **Spatial correction** → — Layer 3, on the residual

The loop was built so that a new destination in the transfer matrix is an addition, not a rewrite.

**Key point** With that one addition, harvest intensity and rotation length join the climate pathway as scenario axes, and the framework begins to answer management questions.

# Conclusion

1. Carbon, habitat and disaster were reconstructed for the whole of Korea on a 1 km grid, each cell keeping its real species mixture rather than a single dominant species.

2. Published equation forms were re-estimated on the basis they are actually read on, which closed a 2.2-fold overstatement of reconstructed volume against national statistics.

3. An annual loop couples growth, mortality, disturbance, decay and establishment under a closed carbon balance, and reproduces remeasured inventory plots at 0.86 to 0.88.

4. Projected to 2100, the stock peaks in 2079 while the annual sink halves twice and closes around 2080, taking the forest from a sink of 56 to a source of 15 Mt $CO_2$ per year.

5. The interval at 2100 is 177 tC per hectare with a 90 per cent range of 154 to 206, and 96 per cent of that variance is parameter uncertainty, so the next gains lie in growth and mortality, in harvest, and in the spatial correction.

**CRAFT gives them a common grid, and makes that grid move.**

산림경영정보학회
Korean Society of Forest Management and Information
KMU 국민대학교

# Thank you

Heedo Choi[1]* · Jee Soo Park[1] · Chul-Hee Lim[1,2,3]†

[1] Department of Climate Technology Convergence, Kookmin University
[2] Department of Forestry, Environment and Systems, Kookmin University
[3] Department of Forest Resources, Kookmin University

zxsa0716@kookmin.ac.kr

**Acknowledgments**

This research was supported by a National Research Foundation of Korea grant provided by the Ministry of Science and ICT (No. 2022R1C1C1008489), and by the Korea Forest Service Government as "Graduate School Specialized in Carbon Sink".